

Munaza Ashraf

Applied ML Engineer and Researcher
World Models and Physical AI

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Education

- Sep 2020 **B.Sc. (Hons.) Software Engineering**, *University of Engineering and Technology (UET)*
Jun 2024 *Taxila, Pakistan*
- CGPA **3.94/4.0**, **Silver Medalist** (second highest aggregate across all four years).
 - Final Year Project: PsycheSync, an AI mental health and wellness platform built on fine tuned LLMs.

Research Interests

- Areas** World Models, Embodied and Physical AI, World to Policy Distillation, Efficient Multimodal Learning, Robot Learning
- Direction** Building **world models for physical AI** that capture how the world moves and changes, and are small and fast enough to run on real robots. My focus is knowledge distillation: transferring the spatial, temporal, and physical understanding of large multimodal models into compact models and policies that fit tight latency, memory, and compute budgets.

Publications & Preprints

- 2026 **Recovering a Withheld Audio Tokenizer from Its Generator: Reverse Distillation of the MiniMax Music 3 Encoder, In and Out of Distribution**
Under Review Munaza Ashraf, Bghira. *Under review*, 2026. [Code] [Models] [Training Data] [MM3 OOD]
- Recovered the withheld tokenizer of the open 8B MiniMax Music 3 generator by reverse distillation, giving a text only model audio prompting: reference guided generation and continuation.
 - The encoder **transfers out of distribution**: recorded music keeps as much reachable conditioning as generated music (**0.92 vs. 0.91**), and degraded view training restores robustness to lossy audio (**0.82 vs. 0.50**).
 - Released the encoders, code, and MM3 OOD, a new **240** track out of distribution benchmark.

Independent Research & Reproductions

- Aug 2026 **Reimplementation: WPT**
Sep 2026 *World-to-Policy Transfer via Online World Model Distillation*. [Paper] [Code]
- Reimplemented WPT with a frozen pretrained **OccWorld** teacher on nuScenes in place of the paper's Drive-OccWorld, and open sourced the full pipeline: world model rollouts, trajectory proposal, reward scoring, and policy distillation into a compact student.
- May 2026 **Reproduction: HASTE**
Head Wise Adaptive Sparse Attention for Video Diffusion. [Paper]
- Reproduced training free sparse attention for Wan 2.2 and LTX 2.3: temporal mask reuse with error guided, per head top p calibration.
- 2026 **Low Latency Content Safety Classifier**, *Runware*
[Base Model]
- Fine tuned Freepik's EVA based vision transformer for a production moderation taxonomy: graded sexual content, suspected minor sexualisation, graphic violence, and extremist content.
 - Returns per category confidence in **under one second**, running on every generation request.
- 2025 **Independent Benchmark: Diffusion Inference Caching**
2026
- Benchmarked FBCache, TeaCache, MagCache, Cache DiT, and Fast3D Cache across image, video, and 3D models, measuring kernel time and peak VRAM against FID, CLIP score, SSIM, and LPIPS.

- May 2026 **Efficient Fine Tuning and Optimizer Studies**
- Jul 2026
- Studied TREAD token routing with LoRA and context parallelism, and ran controlled optimizer comparisons (SOAP, Muon, AdamW, Lion) for diffusion, LLM, and VLM adapters.
 - Designed segmented gradient checkpointing for Flux and MageFlow, giving a finer memory versus recompute trade off and lower step latency.

Research and Work Experience

- Nov 2024 **Applied ML Engineer, Runware, London, UK (Remote)**
- Present
- Tools:** PyTorch, Diffusers, Transformers, vLLM, Go, CUDA, ROCm, NCCL, FlashAttention, SageAttention, FlashInfer, Docker, Prometheus, Vercel
- Integrated, optimized, and served **1,000+** production models across image, video, audio, 3D, diffusion, LLM, and VLM architectures, with up to **2×** faster inference, using model specific schedulers with batching, VRAM aware placement, and hot swapping.
 - Designed distributed and serverless inference across NVIDIA Hopper, Blackwell, Ada, and Ampere GPUs (H100, H200, B300, L40S, RTX PRO 6000, RTX 4090, RTX 3090) and AMD ROCm, with context parallelism and NCCL communication overlap.
 - Benchmarked FlashAttention, SageAttention, FlashInfer, and cuDNN SDPA per GPU architecture to route each model to its fastest attention backend, combined with diffusion and KV caching and FP8, FP4, INT8, and INT4 quantization.
 - Designed and evaluated **teacher–student knowledge distillation** for compact, lower latency model variants, and deployed LoRAs trained once on a base model across its distilled and turbo variants by skipping or bridging the layers with the highest variance or modulation shift.
 - Built the LoRA training platform and Go microservices over vLLM with LLMOps.
 - Served high volume enterprise workloads for Higgsfield, Freepik, Wix, PixVerse, Civitai, and the LTX team, with integrations deployed on Vercel.
- Sep 2024 **Founding Engineer, MedTalk, Faz Australia, Australia (Remote)**
- Jul 2025
- Tools:** Whisper, AssemblyAI, MedPaLM, LLaMA, RAG, Pinecone, WebSockets, AWS, RunPod
- Founding engineer for MedTalk, a real time clinical AI platform used by **50,000+** clinicians; owned ML, backend, and production ML infrastructure.
 - Curated clinical training datasets and fine tuned speech, medical NLP, and language models including MedPaLM for structured clinical note generation.
 - Cut post VAD streaming speech latency from **~150 ms to ~50 ms P50 (3×** faster) with WebSocket streaming and optimized model serving.
 - Built a RAG based clinical assistant and a retrieval augmented billing code LLM on Pinecone and AWS.
- Jun 2024 **Computer Vision Engineer, VisionRD, Islamabad, Pakistan**
- Nov 2024
- Built a wearable AI assembly copilot for Toyota (head mounted 4K camera, real time error detection, cycle analysis), improving line efficiency by **~35%**.
 - Built worker activity and assembly validation for Thal Engineering and visual defect detection for KIA Motors.
- 2022 **STEM Trainer, Junior Coderz, Islamabad, Pakistan (Remote)**
- 2024
- Taught Scratch, Python, app development, and robotics to students in Canada, the USA, and the UAE, and coached students for USACO, Bebras, and AI projects.
- Mar 2023 **Machine Learning Intern, NESCOM, Islamabad, Pakistan**
- Oct 2023
- Developed a deep learning ad detection and classification system, from data collection to evaluation.
- May 2022 **Deep Learning Intern, Bytewise Limited, Islamabad, Pakistan**
- Sep 2022
- Trained computer vision and NLP models in TensorFlow and PyTorch with custom datasets and augmentation pipelines.

Selected Projects

- Aug 2026 **Juricratic: AI Litigation Intelligence, Personal Venture, with Connor, juricratic.com**
- Present
- Models legal cases as imperfect information games, simulating outcomes with role bound LLM agents (judge, jury, opposing counsel, client).
- Mar 2026 **Padelytix: Real Time Multi Camera Sports Perception**
- Aug 2026
- Player and ball tracking, clothing based reidentification, court homography, and highlight extraction, robust to occlusion and camera timing mismatch.

- Apr 2025 **Unilever × PSL 2025: Broadcast Logo Detection and Sponsorship Analytics**
Jun 2025
 - Real time YOLOv8 pipeline that detects, tracks, and grades sponsor logo visibility in live cricket, producing automated exposure and ROI reports.

Honors and Awards

- 2024 **Silver Medal**, *UET Taxila*, second highest aggregate in the session across all four years
2022 to 2023 **Laptop Award**, *Prime Minister's Youth Laptop Scheme*, Pakistan
2018 to 2020 **1st Position in Pre Engineering and Full Tuition Waiver with Stipend**, *HSSC*

Technical Community & Leadership

- 2020 **Active Member**, *Google Developer Student Club (GDSC)*, UET Taxila
2024
 - Organised workshops and led hands on sessions on TensorFlow, Firebase, and Flutter.

2020 **Active Member**, *Microsoft Learn Student Ambassadors (MLSA)*
2024
 - Co organised talks and study jams on Azure and ML, and mentored peers on deploying ML projects.

2020 **Event Host**, *IEEE, UET Taxila Chapter*
2024
 - Hosted seminars and panels connecting students with industry professionals in AI and software.

2020 **General Secretary**, *Umeed-e-Subh*, UET Taxila
2024
 - Student run welfare community that established a school for underprivileged children across from UET Taxila.
 - Helped run daily classes and an evening IT drive teaching graphic design, basic coding, and video editing so children could earn while still in school.

Relevant Coursework

- Mathematics** Calculus, Linear Algebra and Differential Equations, Probability and Statistics
ML & Vision Machine Learning, Neural Networks, Computer Vision, Digital Image Processing
Systems Data Structures and Algorithms, Operating Systems, Databases, Cloud Computing

Technical Skills

- Programming** Python, C++, CUDA, Go, SQL, Unix Shell
ML PyTorch, TensorFlow, Diffusers, Transformers, PEFT, TorchAO, LoRA and QLoRA, Diffusion and Flow Models, LLMs, VLMs, Knowledge Distillation
Train & Infer FSDP, DTensor, Tensor and Context Parallelism, Quantization, Mixed Precision, FlashAttention, SageAttention, Sparse Attention, KV and Diffusion Caching, PyTorch Compilation
Systems NVIDIA Hopper, Blackwell, Ada, Ampere; AMD ROCm; Triton, CUTLASS, cuDNN, NCCL, vLLM, Docker, Kubernetes, AWS, RunPod
Vision YOLO, SAM, Detectron2, Tracking, Reidentification, Pose Estimation
Evaluation Nsight Systems and Compute, Latency and Throughput Benchmarking, FID, CLIP Score, SSIM, LPIPS

References

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